A JACOBSON SPACE WITH FINITELY MANY CLOSED POINTS IS DISCRETE

THE STACKS PROJECT AUTHORS

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Lemma 0.1. *A finite Jacobson space is discrete. A Jacobson space with finitely many closed points is discrete.*

Proof. If X is finite then the set $X_0 \subset X$ of closed points is finite. Assume X_0 is is finite and X is Jacobson. Then $\overline{X_0} = X$ by the Jacobson property. Now $X_0 = \{x_1, \ldots, x_n\} = \bigcup_{i=1,\ldots,n} \{x_i\}$ is a finite union of closed sets, hence closed, so $X = \overline{X_0} = X_0$. Every point is closed, and by finiteness, every point is open. $\square$

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